

FiLM-UNet Meets CBAM-ResNet: A lightweight Pipeline for the UUSIC Challenge

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1 Introduction

Deep learning has significantly advanced medical image analysis in recent years, particularly in segmentation and classification across several imaging modalities. Among the medical acquisition modalities, ultrasound (US) stands out as one of the most common due to its low cost, and real-time acquisition. At the same time, US poses distinctive challenges for automatic analysis: speckle noise and low signal-to-noise ratios, intensity nonuniformities, weak or ambiguous anatomical boundaries, and operator and scanner dependence that induce domain shifts. Over the past year, the community tried to mitigate these difficulties through the release of larger, more heterogeneous US datasets and the organization of benchmarks and challenges.

Within this context, the *Universal UltraSound Image Challenge* (UUSIC) has been conceived as a multi-task benchmark aimed at developing a single model capable of performing classification, and segmentation across multiple organs and diseases. This technical report presents our solution to UUSIC. Our contributions are threefold. First, we demonstrate that Feature-wise Linear (FiLM) [7] is an effective and parameter-efficient mechanism to adapt a single 2D U-Net [8] to segment multiple pathologies targeting different organs. Second, we investigate a lightweight ResNet-18-based classifier improved with Convolutional Block Attention Module (CBAM) [12] to perform classification, and the effective integration of its extracted features to boost the segmentation. Third, we explore using the UNet encoder’s multi-scale features themselves to perform classification, yielding a compact, end-to-end multi-task model.

2 Model: Multi-Task FiLM-UNet with CBAM-ResNet

2.1 FiLM-UNet for Segmentation

The segmentation backbone is a five-scale 2D U-Net operating on 512×512 inputs. Each convolutional block consists of a 3×3 convolution with stride 1 and padding 1, followed by instance normalization and a LeakyReLU activation.

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Downsampling is performed by 2×2 max pooling at each scale and mirrored by symmetric upsampling in the decoder via transposed convolutions with kernel size 2 and stride 2. Skip connections concatenate encoder features with decoder activations at matching resolutions, and a final 1×1 convolution maps the last decoder features to the desired number of segmentation logits.

Organ-aware conditioning is implemented with FiLM [7] layers applied after each convolutional block in both encoder and decoder. Given a feature tensor $x \in \mathbb{R}^{B \times C \times H \times W}$ and an organ identifier $o \in \{0, \dots, N_{\text{org}} - 1\}$, FiLM learns an affine transformation:

$$y = \gamma(o) \odot x + \beta(o),$$

where $(\beta, \gamma) \in \mathbb{R}^{B \times C}$ are learned by embedding the organ ID into a 64-dimensional vector and passing it through a small MLP that outputs $2C$ parameters. As suggested by the original paper, we initialize FiLM to the identity mapping (i.e., $\gamma = 1$ and $\beta = 0$), which stabilizes early training.

2.2 Classification Branch

For image-level classification, we tried two different strategies, one leveraging the U-Net encoder’s features and a final head which process them, and another using a ResNet-based classification branch, then integrated with the U-Net decoder to improve also the segmentation results.

Leveraging Encoder Features for Classification. For the first strategy, we explored two different variants built on top of the segmentation encoder. In the first variant, we used the deepest encoder features by flattening them into a single dimension. This tensor is passed to a layer normalization and used to predict the class with a stack of two linear layers with GELU activation. In the second version, we tried a multi-scale approach by globally average pooling each channel of the last three encoder stages’ features, and concatenating them. This embedding is mapped to classification output with a two-layer MLP with ReLU.

Classification Branch with CBAM Integration. Our second approach uses a dedicated ResNet-18 network for classification. To enhance feature extraction and address the issue of overfitting markers common in ultrasound image classification, we integrate a Convolutional Block Attention Module (CBAM) into the architecture. CBAM sequentially generates attention maps across both channel and spatial dimensions. This dual mechanism guides the network to focus on the most informative features and their optimal spatial locations.

2.3 Controlled Feature Fusion via Tanh Gating

Our multi-task learning approach integrates features from both the classification and segmentation branches to enhance overall performance. The U-Net is improved by incorporating rich, high-level semantic information from the ResNet-18 encoder. The fusion process begins by aligning the output feature maps of the ResNet-18 encoder to the U-Net’s bottleneck layer, through a linear projection, which matches the dimensions of the classification features to the channel

dimensions of the U-Net’s bottleneck embeddings. A crucial component of this fusion is a learnable gating parameter, denoted as α , which is initialized to a tensor of zeros. A hyperbolic tangent ($\tanh$) function is applied to this parameter to produce the final gate values. As the model learns, the values of α are gradually adjusted, allowing the segmentation model to leverage valuable semantic context without compromising its ability to perform accurate pixel-level localization. This controlled fusion process is defined by the following equation:

$$F_{\text{fused}} = F_{\text{UNet}} + \tanh(\alpha) \odot F_{\text{ResNet}} \quad (1)$$

where F_{fused} is the final fused feature, F_{UNet} represents the U-Net’s bottleneck embeddings, F_{ResNet} are the linearly projected classification features, and α is the learnable gating parameter. The operator $\odot$ denotes element-wise multiplication.

3 Methodology

3.1 Dataset and data preprocessing

The ultrasound dataset provided by the challenge combines public and private datasets spanning classification and segmentation tasks, i.e breast nodules classification and segmentation (BUSI, BUSIS [13], and BUS-BRA [2]), thyroid nodules segmentation (DDTI), kidney contour delineation (KidneyUS [10]), fetal head contour delineation (Fetal Head Circumference [11]), cardiac chamber segmentation (CAMUS [4]), Appendix [6] and Fatty-Liver [1] classification. The dataset has a total of 8736 images (7,448 public, and 1,248 private), 6,614 with segmentation masks and 4,945 with classification labels. To standardize field-of-view and remove machine-specific overlays, we applied a deterministic cropping pipeline based on connected component analysis. For each image, we first built a binary mask of “black” pixels (intensity 0, allowing a small tolerance) and extracted the largest component that touches the image border, which captures the non-informative frame/regions outside the ultrasound sector. We then negated this mask and selected the largest remaining component, corresponding to the actual echogenic sector, and used its tight bounding box to crop both the image and, when present, the corresponding annotation masks. This procedure succeeded on all samples except two, which were flagged for manual review. Prior to model input, images underwent intensity normalization and during training, we applied on-the-fly data augmentation.

3.2 Implementation Details

The training of our multi-task model is a sequential process, with each branch trained in a distinct phase. The architecture was implemented using Torch 2.7.1, and all models were trained on an NVIDIA L40S GPU with 48GB of memory.

Training of the Classification Branch. The ResNet-18 classification branch is trained first in a two-phase process: a general pre-training followed by fine-tuning. Throughout both phases, we use the AdamW optimizer with a cosine

annealing learning rate scheduler. The initial phase consists of a general pre-training for 20 epochs on the public datasets, to perform classification and bounding box regression. The optimization is driven by a multi-objective loss function combining Focal Loss and standard Cross-Entropy Loss for classification, and a Smooth L1 loss for bounding box regression with an initial learning rate of 1×10^{-4} . Then the classification branch undergoes a fine-tuning phase for approximately 5 epochs on the smaller, private part of the dataset. The training procedure focuses solely on classification, using the Focal Loss and Cross-Entropy Loss, with an initial learning rate of 1×10^{-5} . This lower learning rate, combined with the cosine scheduler, ensures that the model fine-tunes its parameters without drastically altering the robust features learned during the pre-training.

Training of the Segmentation Branch. We train the UNet for segmentation from scratch, freezing the classification branch, which output features are fused in the bottleneck using the tanh gating mechanism (Sec. 2.3 for details). The UNet is trained using a multi-objective loss function that includes Binary Cross-Entropy (BCE) Loss and Dice Loss, using the SGD optimizer with an initial learning rate of 2×10^{-2} and a LambdaLR scheduler.

3.3 Postprocessing Strategy

We apply a connected-components post-processing to binary masks. Specifically, we first smooth the mask with a small morphological closing and opening (3×3 elliptical kernels) to suppress speckle and thin artifacts. We then run 8-connected components analysis, retain the largest component and also keep the second largest only if its area is at least 30% of the largest (to preserve plausible bi-lobed shapes), and finally fill small internal holes ($\leq 0.3\%$ of the ROI). Instead, for thyroid, which have more heterogeneous and less precise output masks, we use a simplified rule that removes components smaller than 0.1% area of the whole image, suppressing residual speckle while preserving the main structure.

4 Results and Conclusion

From the results presented in Table 1, it is evident that using a U-Net backbone is not a suitable approach for this classification task. This reinforces the hypothesis that the primary objective of segmentation, for which U-Net was designed, is in direct conflict with the demands of a classification objective. While the results with the ResNet-18 backbone show a significant improvement, demonstrating that a dedicated classification architecture is far more effective. A major challenge in classifying ultrasound images is preventing the model from overfitting to irrelevant markers or artifacts from different acquisition machines, rather than focusing on the anatomical features. A qualitative evaluation suggests that the CBAM [12] module is highly effective at helping the model focus on relevant features, a finding that is also supported by the quantitative results from the table. Furthermore, the data reveals a true correlation in performance only when there is a sufficient amount of public and high-quality data available, as is the case for the breast nodules task.

Model	CBAM	FiLM	Appendix	Breast	Breast Lum.	Liver	Overall
U-Net [8]	✗	✗	47.3%	51.2%	17.3%	46.7%	40.6%
	✗	✓	48.8%	46.5%	26.9%	42.4%	41.2%
ResNet-18 [3]	✗	✗	61.2%	62.0%	9.3%	52.4%	46.2%
	✗	✓	42.7%	67.8%	61.9%	51.8%	56.05%
	✓	✗	61.2%	68.7%	59.3%	53.4%	60.7%
	✓	✓	61.2%	65.3%	65.7%	47.1%	59.8%

Table 1: Performance of different classification strategies, on private and unseen dataset during pre-training (breast luminal performances are negligible since they are not directly trained due to lack of publicly available dataset).

Configuration			DSC/NSD (%)					Average (%)	
FiLM	Crop	Postproc	Breast	Cardiac	Thyroid	Fetal	Kidney	DSC	NSD
✗	✗	✗	81.1/15.1	81.1/5.2	62.7/6.9	92.0/9.4	81.5/9.4	79.7	9.2
✗	✓	✗	82.4/17.0	81.7/ 6.0	67.1/8.8	91.4/9.7	85.2/10.3	81.6	10.4
✓	✗	✗	83.9/18.0	82.9/5.4	66.2/8.5	93.2/11.2	86.2/11.3	82.5	10.9
✓	✗	✓	84.0/ 18.3	83.0/5.9	67.1/9.0	93.3/11.5	88.6/11.8	83.2	11.3
✓	✓	✗	84.2/18.0	83.3 /5.8	67.2/9.2	93.0/11.3	88.1/12.0	83.1	11.2
✓	✓	✓	85.5 / 18.3	83.2/5.8	67.8 / 9.7	93.7 / 12.3	88.7 / 12.1	83.8	11.6

Table 2: Performance evaluation of different segmentation configurations on the validation dataset.

Regarding the segmentation task, in Table 2, we observe that each of the three complementary choices in the segmentation branch, bring some improvements to the metrics. First, FiLM conditioning improves both DSC and NSD across organs by modulating feature responses to the organ context, which reduces cross-organ confusion and stabilizes boundaries. Second, cropping the input is beneficial, especially for datasets that include broad black borders, on-screen markers, and large extra-anatomical region, because it suppresses machine-specific artifacts and background clutter, allowing the model to focus on the relevant anatomy. Third, the light post-processing step further removes small spurious blobs and enforces anatomically plausible masks, yielding steady NSD improvements by cleaning false positives near edges. The combination of FiLM, cropping, and post-processing delivers the best overall performance in both DSC and NSD, indicating that representation conditioning, input focus, and minimal morphological cleanup are effective.

It is important to note that these findings are particularly valuable for small-sized models. We also explored strategies that involved the use of significantly larger models, such as MedSAM [5] and Med-Gemma [9]. However, we did not find a valuable trade-off between the marginal performance gains and the substantial increase in computational resources, including time and memory. Therefore, the strategies outlined here represent an optimal approach for developing efficient and effective models for the UUSIC challenge’s tasks.

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